Part A

Question 1:

Let

$$\boldsymbol{F} = \mathbb{Z}_2(\omega)$$

Where,

$$\omega^4 = \omega + 1$$

a)
For $z \in F$ we have $z + z = 0$ because $z + z = 2z \equiv 0 \pmod 2$, so $z = -z$ for all $z \in F$.

b)
i.
To find the multiplicative inverse of $1 + \omega^2 \in F$ we by an arbitary element of K.

The given elemnt can be wriiten as,

$a\omega^3 + b\omega^2 + c\omega + d$

Where $a, b, c, d \in \mathbb{Z}_2$, so we need to find a, b, c, d such that,

$$1 = (\omega^2 + 1)(a\omega^3 + b\omega^2 + c\omega + d)$$

$$= a\omega^5 + b\omega^4 + c\omega^3 + d\omega^2 + a\omega^3 + b\omega^2 + c\omega + d$$

Since $\omega^4 = \omega + 1$,

we can write $\omega^5 = \omega\omega^4 = \omega(\omega + 1) = \omega^2 + \omega$, so,

$$= a(\omega^2 + \omega) + b(\omega + 1) + c\omega^3 + d\omega^2 + a\omega^3 + b\omega^2 + c\omega + d$$

$$= (b + d) + (a + b + c)\omega + (a + b + d)\omega^2 + (a + c)\omega^3$$

ii.

This gives us the system of equations,

$$b + d = 1 \in \mathbb{Z}_2$$

$$a + b + c = 0 \in \mathbb{Z}_2$$

$$a + b + d = 0 \in \mathbb{Z}_2$$

$$a + c = 0 \in \mathbb{Z}_2$$

From the last equation we have $c = a$, and substituting this into the second equation gives us $b = 0$, and substituting this into the third equation gives us $d = a$, and substituting this into the first equation gives us $a = 1$.

Hence, the multiplicative inverse of $1 + \omega^2$ is,

$$\omega^3 + \omega + 1$$

c)